Build Status — What Is Done and What Is Not

As of 20 September 2026, end of session. Counts in this document are generated from the repository, not maintained by hand: the block results come from `sim/run_sim.py`, the unknowns from the specification's own register.

At a glance

	Count
Circuit blocks simulated	**24**
Blocks passing every check	**24**
Blocks failing a check	**0**
Research memos	**12**
Documents in the bible	**18**
Unknowns closed	**4**
Unknowns partly answered	**4**
Unknowns still open	**9**
Phase	**2** — schematic capture **complete**: 10 sheets, root wired, project netlist checked

All 24 blocks pass. On 19 September it was 16, after a temperature-corner pass found six blocks that fail cold or hot and turned prose caveats into enforced checks. All six are now answered, and the answers are part specifications rather than widened checks — every one of them is written up in `docs/bom_requirements.md` with the block that demands it, and every failing corner is kept as pinned evidence of *why* the requirement exists.

One of the six was not a part-selection problem at all. The metering unit's coil is copper, copper moves 0.39 %/°C, and no better resistor exists to buy. It needed closed-loop current control, which put a 50 mΩ shunt and a sense amplifier on ECU pin 88's return — new schematic content, not a BOM line. See §1.7.

Schematic capture has started; no PCB exists and no firmware has been written. The KiCad project is in `hw/`, with all ten hierarchical sheets created and the first — `power_input` — drawn and netlist-checked. See §7.

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1. Circuit blocks — simulated and passing

Each block is a SPICE netlist plus a set of falsifiable checks. A block "passes" when every claim it makes about itself holds. These are design verification, not layout.

1.1 Power chain — 9 blocks

Block	What it covers	Status
`emi_filter`	Two-stage LC, CISPR 25 conducted, differential mode	Passing
`reverse_battery`	Ideal-diode P-FET vs series Schottky	Passing
`transient_clamp`	ISO 7637-2 pulse 2a, +112 V / 2 Ω / 50 µs	Passing
`load_dump`	ISO 7637-2 pulse 5b, 40 V / 0.5 Ω / 400 ms	**FAILS — deliberately**
`buck_preregulator`	6–40 V to 5 V at 400 kHz	Passing
`sensor_rail`	5 V sensor distribution, one PTC per group	**FAILS — cold corner**
`mcu_pdn`	3V3 decoupling impedance, 100 mΩ target	**FAILS — cold corner**
`supervisor`	Watchdog, brownout, fail-safe gate kill path	Passing
`negative_pulses`	ISO 7637-2 pulses 1 and 3a, negative excursions	**FAILS — real gap**

Why `load_dump` fails, and why that is correct. The TVS chosen against pulse 2a absorbs 46 J at 116 W under pulse 5b — roughly 100× both its single-pulse energy rating and its continuous dissipation rating. The failure is the deliverable: it says the part selection is wrong, not that the simulation is. The fix is a higher standoff voltage so the TVS stays off during a normal clamped load dump, rather than a larger part to absorb it. **Blocked on U16** — whether the alternator is suppressed at all decides which pulse applies.

1.2 Sensor front-ends — 7 blocks

Block	What it covers	Status
`sensor_ratiometric`	Pins 41/35/80/37, 0.5–4.5 V into a 3.3 V ADC	Passing
`sensor_differential`	Shared sensor ground on pin 34, single-ended vs differential	Passing
`ntc_frontend`	Pin 79 charge-air temperature, divider vs current source	Passing
`battery_sense`	Pins 04/06, 6–40 V rail into a 3.3 V ADC	Passing
`discrete_input`	Pins 20/24/71, active-high switched battery	Passing
`trip_module_sense`	Trip-module state, supervised three-state loop	Passing
`cam_frontend`	Pin 46 cam Hall, 5 V swing divided into the 3.3 V domain	Passing

`ntc_frontend` simulates **both** candidate topologies because **U2** is unresolved. One resistance measurement at the machine picks between them; the front-end is designed so that measurement selects a populate option rather than unblocking a redesign.

1.3 Speed and position — 1 block

Block	What it covers	Status
`vr_conditioner`	Crank VR sensor, pins 52/74/30	Passing

1.4 Injection — 3 blocks

Block	What it covers	Status
`injector_boost`	Why the boost stage exists, pins 03/05	**FAILS — both corners**
`injector_turnoff`	Turn-off recirculation into the boost rail	Passing
`boost_converter`	Injector rail reservoir sizing	Passing

`boost_converter` passes but carries a recorded gap: it has **no bleed path.** Its charge source regulates the rail from below only. Harmless while current flowed outward; `injector_turnoff` now pushes energy back in, and a hold-current cutoff would climb the rail about 2 V per event with no way down.

1.5 Actuators — 3 blocks

Block	What it covers	Status
`metering_unit_pwm`	Pin 88, low-side PWM into the fuel metering solenoid	**FAILS — both corners**
`relay_driver`	Pins 50/69, low-side FET with flyback	Passing
`egr_hbridge`	Pins 59/81, positional actuator, feedback on 37	**FAILS — shoot-through**

1.6 Communications — 1 block

Block	What it covers	Status
`can_termination`	Split vs single termination, J1939 250 kbit/s	**FAILS — tempco**

1.7 The six corners that were failing, and what each one bought

All six are closed. None was closed by widening a check: each is answered by specifying a part, with the failing corner kept as pinned evidence. The requirements live in `docs/bom_requirements.md` under the tags below.

Block	Was	Answer	Tag
`can_termination`	123.95 Ω at 200 ppm/°C vs J1939's own 120 Ω	thin-film, 0.1 %, 50 ppm/°C → 121.10 Ω at stacked worst case	`CAN-TERM`
`sensor_rail`	4.15 A at cold + −30 % vs a 3.0 A polyfuse ceiling	PTC R25 2.0 → 3.0 Ω, solved against the ceiling → 2.88 A	`SENSOR-PTC`
`mcu_pdn`	197 mΩ at a wet electrolytic's cold ESR vs 100 mΩ	polymer bulk, ESR **band** 20–50 mΩ → 60 / 92 mΩ	`PDN-BULK`
`metering_unit_pwm`	0.862 A cold / 0.462 A hot vs a 0.675 A setpoint	closed-loop current control → 0.679 / 0.683 A	`MU-ISENSE`
`negative_pulses`	VIN −0.528 V vs the LM5164's −0.3 V	active clamp → −0.068 V	`NEG-CLAMP`

Block	Was	Answer	Tag
`injector_boost`	ratio missed 11.6× at both corners	11.6× was this block's *nominal*, never a requirement	—

Three of those were more than part swaps, and they are the ones worth reading.

The metering unit needed an architecture change. Over −40…+125 °C the coil's resistance goes 7.39 → 13.82 Ω and at fixed duty the current follows it, ±30 %. A hysteretic current regulator holds 0.679 / 0.679 / 0.683 A at −40 / 25 / +125 °C. Closed loop only works while the regulator has duty left: at the hot corner the coil can draw at most 0.977 A at 100 % duty, so the setpoint sits at 69 % of the ceiling. That bound is checked, because a regulator out of headroom degrades silently back to open-loop.

The negative clamp needed a FET, for a structural reason. A Schottky's drop at the 15 A pulse 1 delivers decomposes as 0.367 V junction + 0.180 V bulk, and only the bulk term falls when parts are paralleled — four of them still leave 0.375 V. The rating is really a demand for ≤ 20 mΩ effective clamp impedance *including the junction*, which no junction device meets and a 5 mΩ FET clears four times over. Still open but now bounded: the comparator's propagation delay is unmodelled, so the question went from *"is a 2 ms excursion acceptable"* to *"is a sub-microsecond one"*.

The PDN requirement had to be a band, not a ceiling. Bulk ESR is also what damps the regulator-against-bulk resonance, so at 20 mΩ the peak climbs back to 92 mΩ. "ESR ≤ 50 mΩ" would have been satisfied by a 5 mΩ part that fails the block.

A near-miss worth recording. The active-clamp branch first shared the battery node with the unprotected one, and at 13 mΩ to ground it shunted the fault for that branch too — reporting −0.177 V where the true Schottky-only figure is −0.528 V, i.e. reporting the problem as already solved. Caught because the clamp diode's own ammeter read 13 mA where 15 A was expected.

Two stale checks were also found. `negative_pulses` still demanded 0.707 J from the TVS, from before the clamp Schottky existed — the TVS no longer conducts on negative pulses at all, and the requirement moved onto the part that now carries it (13.58 A peak, 14.5 mJ). And a burst-accumulation check conflated first-pulse settling with accumulation; repeats 2–5 are identical to the printed precision.

One block passes in a way worth reading. `emi_filter` was not failed at its corner, because its −40 dB gate is documented in its own final check as being of unknown adequacy pending measured hardware emissions.

Failing a corner against an admittedly arbitrary number manufactures a verdict. The X7R −15 % corner is recorded as regression pins with a note that the value sits below the nominal gate.

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2. Circuits not yet built at all

These have no netlist, no schematic and no checks. They are the real gaps.

Missing circuit	Pins	Why it is not built
Negative-input protection for the buck	pin 21 chain	**Found 19 Sep.** The LM5164's VIN is rated −0.3 V to 100 V; the negative limit is not a mirror of the positive one and nothing was sized against it. The TVS clamps correctly to −38 V, which is still two orders of magnitude outside what the buck tolerates. A missing component, not a mis-sized one
Intake throttle driver	6-pin connector	**Blocked on U8.** Reported battery-fed, which answers who supplies it, not who commands it. May be out of scope entirely
Pre/post-heat output	unknown	**Blocked on U14.** Same distinction — battery-fed is not the same as ECU-switched. The DSE4522 enables both at 50 °C and this design has no heater output

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3. Verification not yet done

The blocks above are each simulated in isolation, at one temperature, against the stimuli named. These are the gaps in the *verification*, not in the circuits.

Gap	Consequence
~~No temperature corner anywhere~~	**Done 19 Sep**, −40/+125 °C across all blocks. Six fail at a corner — `emi_filter` and `can_termination` from *ordinary specified* component behaviour, not stacked worst cases. Being converted from prose into enforced checks
~~ISO 7637-2 negative pulses unsimulated~~	**Done 19 Sep.** The TVS is now modelled as the bidirectional part it is, which is what made the question answerable. Result: the buck's VIN sees −40.7 V against a sourced −0.3 V absolute maximum — see the new gap below
Common-mode EMI unmodelled	`emi_filter` covers differential mode only. Most real automotive failures are common mode, and modelling it honestly needs a layout to give the return path
Reverse-battery transient — now load-bearing	The switch model turns off instantly, so it always looks perfect. That was a caveat; since `negative_pulses` it is a design dependency. If the controller opens fast enough the buck is isolated; if not it sees the full clamped negative voltage. No controller part is chosen and no turn-off delay is sourced, so both bookends are simulated and neither is claimed
Watchdog fault-detection time	`supervisor` bounds only what happens *after* reset asserts, not how long detection takes

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4. Documents

Document	State
Reconciled Specification	**Binding.** Rewritten §3 on the fitted controller, §4 gained a hardware fail-safe requirement, register extended to U16

Document	State
Field Handbook	Current
Sensor Reference	Current
Circuit Reference	Generated from the simulations at build time, so it cannot drift
Memos 01–12	Dated records. Superseded conclusions carry banners rather than being rewritten

Memo 01 and Memo 02 are substantially wrong and say so at the top. Memo 01 misidentified the engine and advised discarding the correct search term. Memo 02 analysed a controller that is not fitted. They are kept because the reasoning that produced them is the record of how the error happened.

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5. Unknowns

Closed — 4

#	Question	Answer
U11	Engine identity	`3GK550ETA 4SR1`, app code `GK3.8703`, CPCB IV+ under certificate `P25/2825/24`
U1	ECU connector	Bosch `1 928 405 192 / 194`, **code C** keying
U13	Which controller is fitted	**DSE4522 MKII AMF**, not the KG640C the diagram shows
U12	Does the fuel relay remove power?	**No — it only signals**

Three of these four closed against what this project had reasoned. That is the single most useful thing to know about the desk work: it is productive, and it is not reliable without the machine.

Partly answered — 4

U4 CAN expectations (addresses 234 and 44 confirmed) · **U5** injector drive profile (family envelope only, no part number) · **U8** throttle (supply known, control not) · **U14** pre/post-heat (supply known, control not)

Open — 8

U2 charge-air sensor topology · **U6** pin 09 function · **U7** pins 04/06 sense or feed (no longer blocking: the board uses them as load feeds since 8 Oct) · **U9** catalyst sensor routing · **U10** SENT signal identity · **U15** what `CRS-878` denotes · **U16** alternator rectifier suppressed or not · plus U12's contact-type confirmation, which gates nothing

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6. Decisions open, and blocking

Decision	Blocks	Why it cannot wait
~~Integrated driver IC vs discrete FETs~~	—	**DECIDED 19 Sep 2026, memo 12.** Discrete FETs, keeping the 100 V rail. MC33816 caps at 72 V, L9781 at a reported 80 V, and both are pre-drivers that need external FETs anyway
~~Passive pulldown vs active clamp~~	—	**DECIDED 19 Sep 2026, memo 12.** Active clamp, built discretely, at every gate. The integrated route relocates this problem rather than removing it
MC33816 last-time-buy 06/08/2027	Fallback path only	Needs a distributor conversation, not another search
L9781 datasheet unretrievable	Confirming the decision's runner-up	Five failed fetches across two sessions. Needs FAE or distributor access
Ideal-diode controller for Q1	`power_input` sheet, `reverse_battery`, `negative_pulses`	Its turn-off time is load-bearing and unsourced. `VBAT_REV_GATE` is undriven on the sheet until a part is chosen
Negative-clamp comparator for Q2	`power_input` sheet, `negative_pulses`	Its propagation delay is the last unmodelled term in the negative-pulse chain. `NCLAMP_GATE` is undriven until a part is chosen
Air-intake shutoff	Nothing electrical	Owners' decision. Nothing in this design can stop an oil-fuelled runaway
Trip-module pickup target	Module ordering	Whether it shares our flywheel target. Sharing is cheaper; separate preserves the independence that is the reason for fitting it

7. Phases

Phase	Output	State
0	Reconciled specification	Done
1	Research memos	Done — twelve
1.5	Block-level simulation	**Done — 25 blocks, 25 passing** (power_path added 8 Oct 2026)
2	**KiCad hierarchical schematic capture**	**Complete — 196 components, 113 multi-pin nets, 18 named open ends**
3	4-layer PCB layout, DRC, fab outputs	Not started
4	Firmware skeleton	Not started

Phase 2 — what exists

The project is `hw/ecu25kva.kicad_pro`. Everything in it is generated, and the generators refuse to overwrite a sheet that has content, so Eeschema edits are safe against a re-run.

Sheet	State	Blocks it realises
`power_input`	**Drawn, netlist-checked**	`emi_filter`, `transient_clamp`, `load_dump`, `negative_pulses`, `reverse_battery`
`rails`	**Drawn, netlist-checked**	`buck_preregulator`, `mcu_pdn`, `sensor_rail`

Sheet	State	Blocks it realises
`mcu`	**Drawn, netlist-checked**	`supervisor`, `mcu_pdn`
`injector`	**Drawn, netlist-checked**	`boost_converter`, `injector_boost`, `injector_turnoff`
`metering_egr`	**Half drawn** — metering unit done; EGR blocked on a rail rating	`metering_unit_pwm`, `egr_hbridge`
`sensors_analog`	**Drawn, netlist-checked**	`sensor_ratiometric`, `sensor_differential`, `ntc_frontend`, `battery_sense`
`speed_inputs`	**Drawn, netlist-checked**	`vr_conditioner`, `cam_frontend`
`discrete_io`	**Drawn, netlist-checked**	`discrete_input`, `trip_module_sense`, `relay_driver`
`can`	**Drawn, netlist-checked**	`can_termination`
`connector`	**Drawn, netlist-checked**	— (not a circuit: the harness boundary)

The S32K148 symbol is generated from NXP's own pin table, and no pin number is retyped anywhere in the chain. KiCad 7 ships no S32K symbol. The package pin numbers live only in `S32K148_IO_Signal_Description_Input_Multiplexing.xlsx`, which is not a download but one of eleven workbooks embedded as attachments inside the Reference Manual PDF. That file had been extracted once and not kept, so `docs/pinmap.md` cited a source the repo no longer held. It is now in `refs/`, with checksums, and `hw/extract_pinmux.py` makes the extraction reproducible. Re-extracting confirmed three of pinmap.md's claims independently: 128 port + 16 supply = 144 pins with every package pin accounted for; the six driver-gate pins are all PE=0 PS=0; and exactly four pins on the whole part carry a reset pull, which are precisely the four JTAG/RESET pins pinmap.md reserved.

Three parts have moved from a class to a part number, each against a retrieved datasheet, and all three are in `docs/bom_requirements.md`: `LM5164DDAR` (buck), `TLV76733QWDRBRQ1` (3V3), `TPS3850G33DRCT` (supervisor). Two of them supersede rows in research memo 07, which is now annotated in place.

The buck row is worth reading as a process failure, not just a part change. Memo 07 chose a 60 V part for load-dump margin, correctly, at the time. The input-protection work then raised the TVS standoff and put the pulse 2a clamp at 73.3 V — and `transient_clamp.cir` has carried a *passing* check reading "60 V-class parts are NO LONGER viable downstream" ever since. The contradiction sat between an executable file and a prose table for days, and only the executable one knew. Nothing compares them. It surfaced when someone had to draw the buck and needed a pinout.

Check the netlist, not the plot. Three bugs on the first sheet were invisible on a correctly-plotting page and obvious in the exported netlist: a rotation helper with 90 and 270 swapped, which put the clamp

Schottky's cathode on ground; custom part fields starting at property id 2, which KiCad reserves, silently renaming the first one "Footprint"; and labels offset from their wires for looks, which detaches them silently while the page still reads as connected.

A wire endpoint landing mid-span on another wire does not connect — not in KiCad 7's netlister, and not even with an explicit junction element in the file. Interactively, Eeschema *splits* the underlying wire when you drop a junction on it; a generated file has no such split, so the two cross without meeting. The page plots exactly as intended and the netlist reports every tapped part as unconnected. Found on the rails sheet, where the input capacitor, the UVLO divider and the RON resistor all tapped the middle of one wire and all three came back unconnected. `schlib.Rail` now emits one segment per interval and is the only thing allowed to draw a multi-tap rail.

Drawing the sheets is finding things nothing else was going to find. Six so far, all of the same shape — a document and an executable file disagreeing, with nothing that compares them:

- **No watchdog pin.** The pin map assigned 37 signals and none serviced the external watchdog memo 11's supervisor topology requires. A window watchdog nobody feeds resets the board on a timer. `PTE0` / `PTE1` added.
- **The buck was a 60 V part on a 73.3 V rail.** `transient_clamp.cir` had carried a *passing* check saying 60 V-class parts were no longer viable for days, while memo 07's BOM still named one.
- **The trip sense was a GPIO on a three-state loop.** `trip_module_sense.cir` designs and checks three bands with real margin; a GPIO reads two, which discards exactly the distinction the bleed resistor exists to create. Moved `PTD6` → `PTB0` (ADC0_SE4).
- **The cam front-end put 5 V on a 3.3 V pin.** Spec §4 and pinmap §1.3 both gave pin 46's front-end as "pull-up to `5V_SENSOR` → timer capture", and no `sim/blocks/*.cir` file simulated a cam front-end at all — `vr_conditioner.cir` covers the crank and there was no cam equivalent, so the swing was never checked. `cam_frontend.cir` was written, and one 16 kΩ resistor turns the pull-up into the same 10 k/16 k divider the ratiometric channels already use.
- **A ground-referenced kill clamp cannot service a high-side gate.** Spec §4's fail-safe list names pins 03/05 alongside the four low-side gates, and `supervisor.cir` sized a ground-referenced clamp; pinmap §2 confirmed the reset state of all six pins against NXP's own table. All three agree, and all three are silent on the fact that the source of a high-side N-FET swings to the boost rail. The injector sheet puts the 470 Ω across gate and *source*, and moves the kill path into the driver's shutdown input.

- **The power ground has no connector pin.** `GP3.8703.C4.pdf` shows pin 21 as the battery positive feed and no battery negative anywhere, because it is a field troubleshooting diagram and says so: it omits "power grounds, unused pins and internal-only pins". Every other sheet returns to `GND` — the injector low sides put 18 A into it — and at the harness boundary that net has nowhere to go. It cannot be solved by choosing one of the 50 unknown pins: a wrong ground pin is a harness that has to be remade, not a rework.

A seventh finding came out of the parts, not the pins. Every clamp earlier in this project was chosen by forward drop. The crank input's four clamp diodes have to be chosen by *reverse leakage* instead: Schottky leakage at 125 °C puts ~0.90 V of offset across the comparator input's 4.489 kΩ Thévenin, which is 4.5× the whole ±198 mV hysteresis window — the comparator latches and a running engine has no speed signal. `vr_conditioner.cir` cannot see this: its comparator is an ideal switch with no input network, so no clamp, no leakage and no offset exist in it. Tagged `VR-CLAMP` in `docs/bom_requirements.md`.

The battery rail's rating is now enforced, not narrated. VBAT_PROT reaches 40.0 V for 400 ms on a load dump and 73.3 V for ~50 µs on pulse 2a, and `transient_clamp.cir` checks what that excludes: a 40 V-abs-max integrated driver by 1.8×, with the dump alone leaving it no margin. The practical consequence is that integrated automotive H-bridges and smart switches — a 40–45 V class — are excluded from everything battery-connected here, which is what blocked the EGR half of `metering_egr`.

Two open items were closed by drawing, not by deciding. The injector current-sense placement (`pinmap` §1.6, INFERRED per-bank, "settle when the sheet is drawn") is settled: low-side, ground-referenced, one per bank, because cylinders 1 and 3 fire 240 crank degrees apart and never overlap. And the boost reservoir's missing bleed path — which `injector_turnoff.cir` asked for by name — is now modelled in `boost_converter.cir` as arrangement C and drawn as a 22 kΩ resistor: one hold-cutoff event clears in 21.4 ms against the 26.67 ms between cylinders, for 455 mW burned continuously.

The root sheet is wired and the project has a netlist for the first time. Until 21 September 2026 `hw/ecu25kva.kicad_sch`'s ten sheet symbols carried **no hierarchical pins**, which meant every sheet's own netlist was correct and the project had none: a name on two sheets was two nets and nothing compared them. Each symbol now carries one pin per net its child exports — 162 pins — with a stub and a label, and two labels with the same name are one net.

Exporting it immediately found the second half of the same problem: **`kicad-cli` reported annotation errors**, because each sheet generator allocated `R1`, `C1`, `U1` from its own counter. Ten sheets, ten parts called `R1`. `schlib.REF_BASE` now gives each sheet its own hundred — KiCad's own sheet-number × 100 convention, applied at generation time rather than by a hand pass in Eeschema.

components	**201**, every reference unique
named nets with ≥2 pins	**108** (the four `5V_MAIN` islands merged into one)
named nets with 1 pin	**18**, each a recorded open item
sheet paths	10
`GND` / `3V3_MCU` / `5V_MAIN` / `VBAT_PROT`	139 / 36 / 28 / 12 pins, all spanning the project

Exporting it for the first time found three real defects that ten clean per-sheet netlists could not.

1. **`5V_MAIN` was four nets.** The buck's 5 V output is a board rail — the CAN transceivers' VCC, the five difference amplifiers and the discrete inputs' pull-ups all run off it — and it was a *local* label on every one of those sheets. A local label stops at its sheet's edge, so three of the four copies had no regulator on them, and each sheet's own netlist was perfectly correct. `rails` now exports it.
2. **`3V3_MCU` was three.** Same mechanism, on `can` (both transceivers' VIO) and `speed_inputs` (the comparator's supply, both upper clamp diodes and the bias divider — five pins and no rail).
3. **Five supply pins had no decoupling capacitor.** The difference amplifiers on `sensors_analog` were the only undecoupled supply pins on the board. No simulation block was ever going to say so: `sensor_differential.cir` models an ideal amplifier with no supply pin at all.

All three are now permanent checks rather than one-off observations: one walks every `power_in` pin and asks whether a capacitor sits on its net, and one fails any sheet-local net whose name shadows a project-wide one — the exact shape of defects 1 and 2.

`hw/check_netlist.py` is the gate, and it is the counterpart to `sim/run_sim.py`: that one guards the component *values*, this one guards the *connectivity*. It enforces the same rule — **an exemption is not a pass**. Every single-pin net must be listed with its reason, and a net that stops being single-pin must be removed from the list or the check fails. An open item that quietly closes is as much a drift as one that quietly opens.

The full schematic set plots to `docs/pdf/ECU-Schematics.pdf`, 11 pages.

Phase 2 is done. What remains under these sheets is not schematic work — it is eighteen named open ends, and every one waits on a datasheet, a part number, or the machine:

- **nine** driver / current-sense-amplifier interfaces (requirements drawn, parts not chosen)
- **two** controller gates on `power_input`
- **five** for the EGR bridge, blocked on a gate driver above 73.3 V
- **two** with no part and no pin: `BARO`, and `TRIP_LOOP` — which needs one of the 50 unconfirmed connector pins, as does `GND`

Correction, 21 September 2026 — the connector's part number was in the repo and the sheet said it was not. `hw/connector.kicad_sch` was drawn from `refs/ecu-pinout-extracted.md`, which predates the field visit and still says "a complete pinout requires the connector part number and the OEM pin list". `refs/field-evidence-2026-09-18.md` §2b closed U1 three days earlier, off a photograph of the housing angled to the light: **Bosch `1 928 405 192` / `1 928 405 194`, code C**, `>PA66-GF50<`, EDC17 family — read off the physical part, which beats a catalogue match because it cannot be a mis-identification. **Code C is load-bearing:** Bosch codes these housings mechanically so differently-coded plugs cannot mate. What is still missing is the OEM *pin list*, a different document, and the board-side land pattern — neither number returns a catalogue hit, so the footprint has to come through a Bosch distributor. Same shape as the six findings above, and this one was mine: reading the older reference and not the newer one.

The last two placeholder symbols are gone, 22 September 2026. The crank comparator is TLV3201-Q1 — the clamp's comparator, which meets every line of `VR-COMP`. The five difference amplifiers are **INA592** at their native G = ½: the requirement was ≥ 60 dB CMRR and the drawn discrete network could only reach ~48 dB; the on-die network gives 88 dB minimum. Putting that real figure into `sensor_differential.cir` in place of an inferred 80 dB raised the differential front-end's advantage from 90× to **394×**, referred to the sensor. INA592 is not AEC-Q100; no qualified G = ½ difference amplifier was found.

Baro drawn, 22 September 2026 — 2 → 1 open end. Spec §7 deviation 2 adds it deliberately; NXP MPXHZ6115A from a retrieved datasheet, on 5V_MAIN so the existing `SENSOR_5V_MON` channel makes it ratiometric. Its datasheet does not state AEC-Q100; Infineon's KP236 is listed qualified but its datasheet would not resolve, and it is a footprint change — so that is recorded, not guessed.

The one open end left is `TRIP_LOOP`, and no part closes it. It needs a connector pin *and* a harness wire — it is a new signal, not on the OEM diagram — which is field-brief task 5.2.

Connector decided, 22 September 2026: the OEM Bosch 94-way. Spec §7 deviation 1 had said the ECU would use functionally-split sealed connectors with an adapter harness; `hw/connector.kicad_sch` drew the OEM 94-way on the board. The owner decided for the connector described in the field evidence — Bosch `1 928 405 192` / `194`, **code C** — so the sheet stands as drawn and the spec's deviation is struck through and marked reversed. No adapter harness. What that makes load-bearing: the board-side land pattern (field-brief task 5.3, via a Bosch distributor), code-C keying, and the loom's own answers to task 5.1 (power ground) and 5.2 (`TRIP_LOOP`).

The EGR bridge is drawn, 22 September 2026 — 7 → 2 open ends. It had been blocked since 20 September on one datasheet: DRV8873-Q1's 40 V VM against a 73.3 V rail. It went discrete — four 100 V FETs, two **AUIRS2184S** — and the decode `EGR-DRIVER` demands comes from the driver's structure, not a logic gate: one input per leg, so each leg's high and low are exclusive by construction and the 270 A state becomes a high-side brake. The same family as the injector drivers, chosen the opposite way for the opposite reason. **Coast is asserted by copper** on a shared, pulled-down shutdown, which costs one MCU pin, `PTB10`.

The negative clamp latched, and a passing check hid it — 22 September 2026, 8 → 7 open ends. Choosing its comparator meant setting real thresholds, and the model's were wrong: engage near +0.08 V, release above +0.12 V. At the end of every negative pulse the returning source drove 1.35 A back through the still-on clamp — 6.7 mV across 5 mΩ, never reaching release — so it **latched on and shorted the source**. The check measured only the clamp's minimum and said RESOLVED; a recovery check failed on both pulses. It is a low-side ideal diode and now releases above −6.8 mV, below zero even with the comparator's full offset.

Two more numbers on that block were artifacts. With the ~100 ns comparator-and-driver delay in, the default timestep let ngspice commit the hysteretic switch on a rejected trial step: the "−0.068 V" minimum was really **−0.177 V** — still inside the −0.3 V rating by 1.70×, which answers the question this block had carried since 20 September. And the pinned 4.74 A clamp-diode peak and a 15.8 mV "settling" difference on pulse 3a were coarse-step overshoot, confirmed on the unchanged original netlist: 4.114 A fine, equal to the flat top and to (206.5 − 0.4) / 50 Ω.

Choosing the clamp comparator found two under-rated parts and a drawn FET of the wrong polarity, 22 September 2026. `NEG-CLAMP` specified its FET at **40 V** and the backstop Schottky was **60 V** — both with a terminal on `VBAT_PROT`, both sized for the negative pulses they catch, neither checked against pulse 2a driving the *same* node to **+73.3 V**. `transient_clamp.cir` carried "every part on VBAT_PROT must clear

73.3 V" as a principle, not a list; it is now a **table**, every part on the rail with its rating and margin, and a part below the line fails. Both are 100 V class now.

And the reverse-battery FET: the sheet drew an **N-FET** with an undriven controller gate, while `reverse_battery.cir` has always modelled a **P-FET**. The node swings +73.3 V to −47.8 V and no controller IC checked spans it — LM74700-Q1 stops at +65 V, LM5050-1-Q1 at −0.3 V. A P-FET with a passive gate network *is* the block's switch model and has no IC ratings to violate. It gives up ideal-diode hold-up on a *shorted*-input dropout, which the spec does not require; that is on the page.

Current sense chosen, 22 September 2026: INA181A1-Q1 on all three channels, 12 → 9 open ends. One part and one gain — 20 — set for the injector, so the 18 A peak reads 1.80 V and a fault to 32.8 A is still measured rather than clipped. Supplied from 3V3_MCU, which makes its output ADC-safe by construction and removes the need for a clamp. The metering FET was also redrawn: it had been placed at `rot=270` since 20 September, which gives diagonal drain and source wires — the same drawing bug the injector sheet had, fixed there first.

Gate drivers chosen, 21 September 2026: AUIRS2181S on all eight gates, and the open-end count fell from 20 to 12. Its first listed application is common-rail injection, and the deciding property is one a parametric search would never filter for: it has **no cross-conduction interlock**. In a half bridge that is the less safe choice; here the high and low switches are in *series* with the coil and must both be on, so UCC27282-Q1 and UCC27712-Q1 — which have interlock — could not fire an injector at all. It also clears the two ratings the obvious 120 V part (UCC27211A-Q1) fails: 4.1 V of headroom on its HB absolute maximum, and a −1 V HS DC minimum that the freewheel diode violates on every hold off-time. The kill path moves to `HIN`, the one ground-referenced point in the high-side path.

The drivers needed a rail the board did not have — 10–20 V — and **it is made from the boost rail** so it survives cranking. The LM5164 already on the board was the obvious candidate and is ruled out by its own VIN absolute maximum of 100 V against a ~102 V rail. A discrete follower with a current limit carries the ~3 mA; if the boost stops, every driver drops into UVLO and every gate turns off.

Choosing the second part found an error in the first sheet it touched. The injector high side was drawn on 21 September with the battery feeding the bank node through a plain **diode-OR**, on the argument that it needed no extra MCU pin. Looking for a gate driver for it showed that it is wrong three ways, and only the third is visible from a datasheet:

1. **It cannot regulate hold.** With the battery on a diode the only switch in the loop is the low side, and opening it sends the coil's current into the 100 V rail — a fast decay against −87 V, which is turn-off, not chopping.
2. **It leaves connector pins 03 and 05 permanently live**, so a harness short to ground draws current whenever the battery is connected.
3. **The gate driver cannot be bootstrapped.** A bootstrap capacitor charges when the switch node goes low; a battery diode holds it at 12.8 V, so it never does.

The replacement is what `injector_turnoff.cir` already modelled and spends a paragraph of its header justifying: **two high-side switches per bank** onto a common node, plus `D_fw` from **ground**. Hold chopping moves to the high side, the node swings to −0.7 V each off-time, and the bootstrap charges. It costs two MCU pins — `PTB5` and `PTA17`, both FTM0 channels, so the whole injector stage stays on one timer and its edges stay phase-locked. The pin map's "six gate pins" section is now eight.

Phase 3 started 21 September 2026 — the first part is chosen. `TPS40210QDGQRQ1`, the injector boost controller, and it was chosen on **one spec that runs backwards from the usual**: tOFF(min) = 200 ns max. A boost needs 94.1% duty at the 6 V cranking corner; at 150 kHz that off-time floor allows 97.0%, and at the 2.2 MHz most wide-input boost controllers in this class use, the same 200 ns caps duty at 56% — below even the 86.6% the rail needs at a *nominal* battery. Switching slower is the requirement, and the part that looks less modern is the one that works.

And it hit the 73.3 V rail rating for the third time. VDD absolute maximum is 52 V. Same shape as DRV8873-Q1's 40 V `VM`, which took that part off the EGR bridge — solvable here only because this pin draws 2.5 mA plus gate charge rather than a motor's current, so a series resistor and a zener hold it inside the rating. All of it is checked as executable arithmetic under `boost_converter`, not as prose. *(Corrected 22 September: the clamp as first drawn — 47 Ω and a 43 V zener checked as an ideal 43.0 V source — gives 61.9 V with a real part. It is now 220 Ω and a 36 V BZG03C36-HM3, 46.6 V worst case. See §7's parts entry.)*

The converter stage is drawn — `injector` moved to A1 to hold it — and the loop compensation is the one thing on it marked as a starting point rather than a result, because a current-mode boost's loop depends on the inductor's real DCR and the capacitor's real ESR and is a bench measurement.

Small-signal parts chosen, 22 September 2026. Choosing them found two errors, both in gate drive.

1. **Every kill-FET gate could see 73.3 V.** `GATE_KILL` is pulled to VBAT_PROT on purpose, so the kill path has a supply during a 3V3 brownout. Nothing on the net draws DC, so the net follows the rail up to the pulse-2a peak. That is 3.7× a small-signal FET's ±20 V Vgs rating. A BZX84-C12-Q now clamps the net at 12.7 V. See `GATE-KILL-CLAMP`.
2. **The relay FETs' gates were driven from a 3.3 V pin.** The schematic noted them "fully enhanced at 3.0 V", and no 100 V automotive FET checked is specified below Vgs = 4.5 V. An SN74AHCT1G125-Q1 on 5V_MAIN now drives each gate. See `RELAY-GATE`.

Parts chosen: 2N7002BK for the nine kill FETs and the reset inverter, PMV280ENEA for the relay FETs. The same pass fixed a diagonal wire from VBAT_PROT to each relay flyback diode, a drawing bug present since the sheet was first drawn.

Power FETs chosen, 22 September 2026: four part numbers for all sixteen. SQD50P08-25L is the reverse-battery P-FET. IPD90N10S4-06 ×8 covers the negative clamp, metering, the EGR bridge and the injector battery switches. SQM85N15-19 ×5 covers the injector boost-side and low-side switches. DMN15H310SK3 is the boost switch, the one part that needs its Rds(on) specified at a 4 V gate. Two things the choice changed:

- **The symbols.** Every FET had been drawn as `Q_NMOS_GSD`, which is the SOT-23 pin order. DPAK and D²PAK are G-D-S with the drain on the tab. With the TO-252 footprint on the old symbol, the source would have gone to the tab. The power FETs now use `Q_NMOS_GDS` and `Q_PMOS_GDS`, which have the same pin positions and different numbers. The netlist shows every drain on pin 2.
- **Checks that had used placeholders.** The bootstrap droop assumed 60 nC; the chosen part's maximum is 120 nC, which doubles the droop to 126 mV (1.1% of the bootstrap supply). The boost's cranking duty had ignored the switch; with its drop at a hot junction it needs 94.8% of the 97.0% available. The gate pulldowns were sized for Crss = 500 pF; the worst chosen part is 300 pF max.

Diodes chosen, 22 September 2026, with one design error found and one class left open.

- **The boost controller's VDD clamp did not clamp.** It had been checked as an ideal 43.0 V source behind 47 Ω. The real 43 V part is 40–46 V with +0.12 %/K and 45 Ω of impedance, which puts **61.9 V** on a 52 V pin at 125 °C. Its 40 V minimum also conducts on the load dump it was chosen to avoid. It is now 220 Ω and a BZG03C36-HM3: 46.6 V worst case, and 33 mA on the dump. The chosen boost switch's small gate charge is what makes 220 Ω possible. A 39 V zener was tried first, and the `transient_clamp` rail table rejected it at 1.05×. See `BOOST-VDD-CLAMP`.

- **Three stock footprints number their pads against the symbol.** TO-277A (the backstop Schottky) and SOT-23 zeners have the cathode on pad 3. `hw/gen_footprints.py` now derives renumbered copies into `hw/lib/ecu25kva.pretty`. The GATE_KILL zener, footprinted in the previous commit with the stock SOT-23, had been reversed.
- **LOWV-CLAMP is open**, covering the 15 input clamps at 3.0 V and 3.3 V. A sub-5 V zener at tens of µA leaks enough to matter, and the model's sharp-knee zener hides it.
- **Phase-3 notes.** The seven VS-16EDH02HM3 are SMPD, which has no KiCad 7 footprint. The BAV199 pairs are drawn as separate half-parts and need one package per pair.

Transistors and magnetics, 22 September 2026: three more open items, and two of them are design decisions, not part choices.

- **12V_GATE pass transistor.** The DPAK part could not be retrieved. The verified PZTA42-Q in SOT223 would have run 1.01× over its hot rating, so two 3.3 kΩ collector resistors now take half the power. The **105 °C board ambient is an assumption**; the spec states none.
- **POWER-PATH.** The injector and EGR current all flows through pin 21, the EMI filter and a reverse FET sized for 5 A. Pins 04/06 are separate 10 A battery feeds that are used only as senses. Architecture decision needed.
- **BOOST-SLOPE.** By TI's Equation 9, the drawn boost goes subharmonic while cranking (82 mΩ against a 44.9 mΩ ceiling). The 330 µH it needs does not exist at a 2.2 A saturation rating, and a smaller inductor makes the ceiling lower.
- **SENSOR-PTC.** A real PTC is specified from 0.40 to 5.00 Ω, not "3.0 Ω ±30%", and its hold current halves at 85 °C.
- The netlist gate now counts the **power ground** as an open item instead of passing it silently: GND reaches the connector only through sensor-ground pins.

Every chip resistor and capacitor is packaged, 22 September 2026. 188 of them, sized by `hw/assign_passives.py` from the voltage across each part and its continuous power at a 105 °C board. Ratings come from Vishay's CRCW e3 and KEMET's automotive X7R tables, and the netlist gate now fails on any R or C left without a footprint. See `PASSIVE-PACKAGES`. The pass found that 4.7 µF at 100 V does not exist in automotive X7R, so three input capacitors became 2 × 2.2 µF. The EMI model was updated to match.

The 38 parts still without a footprint are all either blocked or chosen separately:

- Blocked on POWER-PATH: L101, L102, F101.
- Blocked on BOOST-SLOPE: L401.

- Blocked on SENSOR-PTC: F201–F203.
- Blocked on LOWV-CLAMP: the 15 zeners at 3.0 and 3.3 V.
- Land pattern needed in phase 3: the seven SMPD diodes, L201 and J1001.
- BAV199 pairs, drawn as separate halves: 5.
- Not yet chosen: C313 (polymer) and C401 (electrolytic). Both chosen 8 October, decisions log §9.

Footprints are the phase-3 input, not a phase-2 gate: 264 of 302 components carry one, and they are exactly the parts with a retrieved datasheet behind them.

POWER-PATH built, 8 October 2026. The injector and EGR loads moved off pin 21. Pin 04 now feeds `VBAT_INJ` (the hold switches and the boost input) and pin 06 feeds `VBAT_EGR` (the EGR bridge). Each has a TVS, an SQM120P10-10m1L reverse P-FET, a Schottky backstop and 4.4 µF of bulk. New block `power_path.cir` (18 checks): EGR stall gives Tj 139 °C at a 105 °C board (155 °C while cranking), and the injector gives Tj 117 °C. The netlist gate checks the split. The result is 333 components, 314 with footprints, and 208 passives packaged. L101, L102 and F101 are unblocked and now need pin 21's reduced load stated before they are sized. Conducted emissions on the two new rails are open. This was done on macOS with KiCad 10 for `kicad-cli` and KiCad 7.0.11's symbol libraries for generation, and ngspice 42.

Pin 21's filter and fuse chosen, 8 October 2026. Pin 21's load is now stated as an upper bound of 2.95 A: the LM5164's 1.75 A peak limit caps the buck's input current, plus the relays, the metering freewheel and the boost VDD feed. The parts are L101 Würth 7447709220 (22 µH, Isat 6.5 A), L102 Würth 744314047 (0.47 µH), and F101 Littelfuse 0437005.WRA (5 A, 63 V, 3.16 A at 105 °C). F101 used to be 20 A. The EMI model now carries the inductors' RDC and self-resonance, and its 150 kHz margin fell to 0.56 dB. The fuse's pulse I^2t is now computed from the simulated waveforms: 16 % of melting on pulse 1, against a 20 % rule of thumb whose pulse-cycling data is still open. 317 of 333 components now have footprints.

SENSOR-PTC built, 8 October 2026. The three PTCs (F201–F203) are gone. Each sensor group now has a TPS7B4253-Q1 tracker following 5V_MAIN, all behind one TPS48110-Q1 overvoltage cut-off. The controller was chosen over the LTC4368 for AEC-Q100: decisions log §7, asked and explained in plain terms. `sensor_rail.cir` was rewritten with pulse 2a through the real pin-21 filter. The trackers' input peaks at 22.3 V (limit 45 V), because the controller's own gate network makes the FET a source follower. The block also found that the clamp node reaches 76.6 V with an unsaturated filter inductor, against 73.3 V in

`transient_clamp.cir`; that is now a check. The result is 353 components, 335 with footprints, and 225 passives packaged. Open: the 47 µF hold-up part, and two land patterns.

BOOST-SLOPE built, 8 October 2026: all three approved redesigns are now in. The TPS40210-Q1 runs from its own output through a 12V_GATE-style follower (34.3–45.0 V); the battery only starts it, through 180 Ω and a Schottky. L401 is 150 µH, Würth 7447709151, not the approved 100 µH. Two questions to the owner settled that (decisions log §8): the owner's 10 × 10 × 4 mm suggestion saturates at 1.4 A, and the 12 × 12 mm 150 µH beats 100 µH on stability and saturation. The sense resistor is 62 mΩ. Equation 9 holds at the worst corner by 1.12×. Building it found that the oscillator's −20 % frequency corner would have failed the 100 µH design outright. The result is 362 components, 345 with footprints, and 230 passives packaged.

The three non-ceramic capacitors chosen, 8 October 2026 (decisions log §9). Each is a polymer type, because a polymer's ESR barely moves in the cold. C313 is a KEMET T599 150 µF with a 22 mΩ resistor in series. No datasheet gives an ESR minimum, so the resistor is `PDN-BULK`'s floor. C211 is a Chemi-Con HXC 47 µF 63 V. C401 is now two HXC 82 µF 80 V cans in series, because no part rated above 100 V publishes a cold ESR; the owner chose that over one 160 V wet can. The injector still reaches 18 A in 45 µs at −40 °C, aged. The sims run the 41 µF stack, and every check uses its worst-case low point, 85.1 V. The 22 kΩ bleed is split 2 × 11 kΩ to balance the cans. The result is 365 components, 351 with footprints, and 232 passives packaged.

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8. Work at the machine

Ordered by what it unblocks.

Re-ordered 21 September 2026, after the schematics were drawn. Two items moved to the top that were not on this list before, because they came out of capture rather than out of a document. The full tasks are `docs/research/08-engine-inspection-brief.md` **Group 5**, written for whoever is at the machine.

Task	Closes	Unblocks
Continuity from every populated cavity to chassis — where is the power ground?	—	**The board, outright.** 94 pins and no identified battery negative; the injector low sides put 18 A into a net with nowhere to land
Cavity census: populated, contact-only, empty	—	A pin **and a wire** for `TRIP_LOOP`, which is not on the OEM diagram at all
Trace who switches the heater feed	**U14**	Whether a heater output is in scope — **can make the board bigger**

Task	Closes	Unblocks
Follow the throttle's 6-pin cable to where it ends	**U8**	Whether a throttle driver exists at all — **can make the board bigger**
Injector part number from the body	**U5**	The boost rail's 100 V target and the 18 A threshold, both currently a family envelope
Alternator rectifier part number, or a scope during a live disconnect under load	**U16**	The whole input-protection topology, and `load_dump` 's failure
Resistance across the charge-air element, cold	**U2**	The `ntc_frontend` populate choice
Crank VR output while cranking, cold *(supervised)*	—	Confirms the ±198 mV hysteresis window against the real sensor, not a 2 V estimate
Contact pitch and row spacing, with calipers	—	Confirms which Bosch family drawing is the right one for the board-side footprint
Relay contact-type check	U12 confirmation	Nothing — recorded for completeness

The two that can grow the board — U14 and U8 — should be answered before layout starts, not after. Everything else on this list changes a value or closes a register entry; those two change the output count.

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9. Resume here — next session

Written at the end of 20 September so the next session does not have to re-derive where things stand. **Nothing is half-finished: the tree is clean, all 24 blocks pass, everything is pushed.**

What closed today

Phase 1.5 is done. All six temperature-corner failures are answered, and the answers are in `docs/bom_requirements.md` rather than in a widened check — see §1.7 for what each one cost and which three were more than part swaps.

Phase 2 started. The KiCad project exists with all ten sheets, the S32K148 symbol is generated from NXP's own pin table, and `power_input` is drawn and netlist-checked.

Next, in order

1. Draw the remaining sheets. The generator and its helper (`hw/schlib.py`) are in place, so each sheet is now a matter of describing its parts and their provenance. Order by what unblocks the most:

- `mcu` — the pin map is done and every other sheet references it
- `rails` — `buck_preregulator` plus the two specified parts (`PDN-BULK` , `SENSOR-PTC`)

- `metering_egr` — carries the new current-sense requirement (`MU-ISENSE`)
- `can` — small and fully specified (`CAN-TERM`)
- then `injector`, `sensors_analog`, `speed_inputs`, `discrete_io`, `connector`

2. Check each sheet's netlist, not its plot. Every bug found on the first sheet was invisible on a page that plotted correctly. `kicad-cli sch export netlist` is the check; KiCad 7's CLI has no ERC subcommand, so this stands in for it.

3. Two parts still need choosing, and both show up as undriven gates on `power_input` (§6): the ideal-diode controller and the negative-clamp comparator. Neither pinout can be drawn without a datasheet.

Needs a person, not an agent

- **MC33816AE last-time-buy 06/08/2027** — a distributor conversation, not another search.
- **L9781 datasheet** — five failed retrievals across three sessions. Needs FAE or distributor access.

At the machine

See §8. **U16 (the alternator rectifier) no longer gates the input-protection topology** — that closed when the TVS standoff and the active clamp were specified — but it still decides which ISO 7637-2 pulse applies, and it is the first thing to settle.

The pattern worth carrying forward

Every wrong number found in this project so far has been found in prose, not in a check. The corner that fails is the one that is written as an enforced check; the corner that is written as a caveat in a comment is the one that gets believed. The same held today at the schematic: three real errors, all invisible on the page, all obvious in the netlist.